

Welcome back! If you found this question useful,
don't forget to vote both the question and the answers up.

[close this message](#)

Dual Problem - Support Vector Machine - Solution by hand

Asked 1 year, 4 months ago Modified 1 year, 4 months ago Viewed 80 times

I am trying to find the solution to the Dual Problem. My data are

$$S_1 = (3, -1), S_2 = (4, 3) \text{ e } S_3 = (1, 1)$$

with

$$y^1 = +1, y^2 = +1 \text{ e } y^3 = -1$$

I solved it using another method and found

$$b = -1.6$$

$$w = (0.8, -0.2)$$

But I couldn't solve the Dual Problem. Please tell me what I am doing wrong:

Method 2: Dual Problem

$$\begin{aligned} J(\alpha) = & \alpha_1 + \alpha_2 + \alpha_3 - \frac{1}{2} [\alpha_1 \alpha_1 y^1 y^1 S_1^T S_1 + \alpha_1 \alpha_2 y^1 y^2 S_1^T S_2 + \alpha_1 \alpha_3 y^1 y^3 S_1^T S_3 + \\ & + \alpha_2 \alpha_1 y^2 y^1 S_2^T S_1 + \alpha_2 \alpha_2 y^2 y^2 S_2^T S_2 + \alpha_2 \alpha_3 y^2 y^3 S_2^T S_3 + \\ & + \alpha_3 \alpha_1 y^3 y^1 S_3^T S_1 + \alpha_3 \alpha_2 y^3 y^2 S_3^T S_2 + \alpha_3 \alpha_3 y^3 y^3 S_3^T S_3] \end{aligned}$$

So I got that

$$\begin{cases} \max_{\alpha} J(\alpha) = \alpha_1 + \alpha_2 + \alpha_3 - \frac{1}{2} [10\alpha_1^2 + 9\alpha_1\alpha_2 - 2\alpha_1\alpha_3 + \\ \quad + 9\alpha_1\alpha_2 + 25\alpha_2^2 - 7\alpha_2\alpha_3 + \\ \quad - 2\alpha_1\alpha_3 - 7\alpha_2\alpha_3 + 2\alpha_3^2] \\ \alpha_1 + \alpha_2 - \alpha_3 = 0 \end{cases}$$

Simplifying

$$\begin{cases} \max_{\alpha} J(\alpha) = \alpha_1 + \alpha_2 + \alpha_3 - 5\alpha_1^2 - \frac{25}{2}\alpha_2^2 - \alpha_3^2 - 9\alpha_1\alpha_2 + 2\alpha_1\alpha_3 + 7\alpha_2\alpha_3 \\ \alpha_1 + \alpha_2 - \alpha_3 = 0 \end{cases}$$

Calculating the partials derivatives:

Welcome back! If you found this question useful,
don't forget to vote both the question and the answers up.

[close this message](#)

$$\frac{\partial J}{\partial \alpha_2} = 1 + 2\alpha_1 + 25\alpha_2 + 7\alpha_3 = 0$$

$$\frac{\partial J}{\partial \alpha_3} = 1 + 2\alpha_1 + 7\alpha_2 - 2\alpha_3 = 0$$

So I ended up with this system

$$\begin{cases} 10\alpha_1 + 9\alpha_2 - 2\alpha_3 = 1 \\ 9\alpha_1 + 25\alpha_2 - 7\alpha_3 = 1 \\ 2\alpha_1 + 7\alpha_2 - 2\alpha_3 = -1 \\ \alpha_1 + \alpha_2 - \alpha_3 = 0 \end{cases}$$

But the solution is a null vector.

[optimization](#) [convex-optimization](#) [machine-learning](#)

[Share](#) [Cite](#) [Edit](#) [Follow](#) [Flag](#)

edited Jul 3, 2022 at 10:34

asked Jul 3, 2022 at 0:26



Trucker

368 2 8

1 Answer

Sorted by: [Highest score \(default\)](#)



Forming the lagrangian

1

$$J(\alpha, \lambda) = \sum_k \alpha_k - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j s_i \cdot s_j + \lambda(\alpha_1 + \alpha_2 - \alpha_3)$$



the stationary conditions are



$$\nabla J = 0 = \begin{cases} -10\alpha_1 - 9\alpha_2 + 2\alpha_3 + \lambda + 1 \\ -9\alpha_1 - 25\alpha_2 + 7\alpha_3 + \lambda + 1 \\ 2\alpha_1 + 7\alpha_2 - 2\alpha_3 - \lambda + 1 \\ \alpha_1 + \alpha_2 - \alpha_3 \end{cases}$$

with solution

$$\alpha_1 = \frac{11}{50}, \alpha_2 = \frac{3}{25}, \alpha_3 = \frac{17}{50}, \lambda = \frac{8}{5}$$

[Share](#) [Cite](#) [Edit](#) [Follow](#) [Flag](#)

answered Jul 3, 2022 at 11:17



Cesareo

31.1k 14 15 45

Welcome back! If you found this question useful,
don't forget to vote both the question and the answers up.

[close this message](#)

so the third equation on the system is

$$2\alpha_1 + 7\alpha_2 - 2\alpha_3 + \lambda = -1$$

– [Trucker](#) Jul 3, 2022 at 18:52 
